

SQA & Testing

Chapter 1

Basic Concepts and Preliminaries

Software quality – definition

IEEE definition

Software quality is:

1. The degree to which a system, component, or process meets specified requirements.
2. The degree to which a system, component, or process meets customer or user needs or expectations.

...Software quality – definition

Other definitions:

“Quality means conformance to requirements” (Crosby, 1979)

“(1) Quality consists of those product features which meet the needs of customers and thereby provide product satisfaction.

(2) Quality consists of freedom from deficiencies” (Juran, 1988).

Conformance to explicitly stated functional and performance requirements, explicitly documented development standards, and implicit characteristics that are expected of all professionally developed software. (Pressman, 2000)

Software quality assurance

The IEEE definition:

Software quality assurance is:

1. A planned and systematic pattern of all actions necessary to provide adequate confidence that an item or product conforms to established technical requirements.
2. 2. A set of activities designed to evaluate the process by which the products are developed or manufactured. Contrast with quality control.

...Software quality assurance

SQA – expanded definition:

SQA is:

A systematic, planned set of actions necessary to provide adequate confidence that the software development process or the maintenance process of a software system product conforms to established functional technical requirements as well as with the managerial requirements of keeping the schedule and operating within the budgetary confines.

SQA Vs. SQC

QC is a set of activities carried out with the main objective of withholding products from shipment if they do not qualify.

QA is meant to minimize the costs of quality by introducing a variety of activities throughout the development and maintenance process in order to prevent the causes of errors, detect them, and correct them in the early stages of development.

QA substantially reduces the rates of non-qualifying products.

QC activities are only a part of the total range of quality assurance activities.

Objectives of SQA activities

The objectives of SQA activities for software development and maintenance are:

- (1) Assuring, with acceptable levels of confidence, conformance to functional technical requirements.
- (2) Assuring, with acceptable levels of confidence, conformance to managerial requirements of scheduling and budgets.
- (3) Initiating and managing activities for the improvement and greater efficiency of software development and SQA activities.

SQA and Software Engineering

- SE is the application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software.
- The characteristics of SE, especially its systematic, disciplined and quantitative approach, make SE a good environment for achieving SQA objectives.
- It is commonly accepted that cooperation between software engineers and the SQA team is the way to achieve efficient and economic development and maintenance activities that, at the same time, assure the quality of the products of these activities.

Role of Testing

- Software quality assessment divide into two categories:
 - Static analysis
 - It examines the code and reasons over all behaviors that might arise during run time
 - Examples: Code review, inspection, and algorithm analysis
 - Dynamic analysis
 - Actual program execution to expose possible program failure
 - One observe some representative program behavior, and reach conclusion about the quality of the system
- Static and Dynamic Analysis are complementary in nature
- Focus is to combines the strengths of both approaches

Failure, Error, Fault and Defect

Failure

- A *failure* is said to occur whenever the external behavior of a system does not conform to that prescribed in the system specification

Error

- An *error* is a state of the system.
- An *error* state could lead to a *failure* in the absence of any corrective action by the system

Fault

- A *fault* is the adjudged cause of an *error*

Defect

- It is synonymous of *fault*
- It a.k.a. *bug*

Causes of software errors

- (1) Faulty definition of requirements
- (2) Client–developer communication failure
- (3) Deliberate deviations from software requirements
- (4) Logical design errors
- (5) Coding errors
- (6) Non-compliance with documentation and coding instructions
- (7) Shortcomings of the testing process
- (8) Procedure errors
- (9) Documentation errors

Verification and Validation

Verification

Evaluation of software system that help in determining whether the product of a given development phase satisfy the requirements established before the start of that phase

Building the product correctly

Validation

Evaluation of software system that help in determining whether the product meets its intended use

Building the correct product

The Objectives of Testing

The stakeholders in a test process are the programmers, the test engineers, the project managers, and the customers.

- It does work (Programmers)
- It does not work (Tester)
- Reduce the risk of failures (Project managers)
- Reduce the cost of testing (Customers)
 - the cost of designing, maintaining, and executing test cases,
 - the cost of analyzing the result of executing each test case,
 - the cost of documenting the test cases, and
 - the cost of actually executing the system and documenting it

Why testing is necessary?

What is a Test-case?

Test-case is a simple pair of < input, expected outcome >.

In stateless systems, where the outcome depends solely on the current input, test cases are very simple in structure.

In state-oriented systems, where the program outcome depends both on the current state of the system and the current input, a test case may consist of a sequence of < input, expected outcome > pairs.

...What is a Test-case?

Examples of basic test cases

In stateless systems,

A program to compute the square root of nonnegative number:

$TB_1: < 0, 0 >$,

$TB_2: < 25, 5 >$,

$TB_3: < 40, 6.3245553 >$,

$TB_4: < 100.5, 10.024968 >$.

...What is a Test-case?

Example of a test case with a sequence of < input, expected outcome >

In state-oriented systems

Automated teller machine (ATM):

TS1:

< check balance, \$500.00 > ,

< withdraw, "amount?" > ,

< \$200.00, "\$200.00" > ,

< check balance, \$300.00 > .

Expected outcome

An outcome of program execution is a complex entity that may include:

- Values produced by the program:
 - Outputs for local observation (integer, text, audio, image)
 - Outputs (messages) for remote storage, manipulation, or observation
- State change:
 - State change of the program
 - State change of the database (due to add, delete, and update operations)
- A sequence or set of values which must be interpreted together for the outcome to be valid.

NB: Ideally, the expected outcome of a test should be computed while designing the test case.

Who does software testing?

Test manager

- manage and control a software test project
- supervise test engineers
- define and specify a test plan

Software Test Engineers and Testers

- define test cases, write test specifications, run tests

Independent Test Group

Development Engineers

Quality Assurance Group and Engineers

- Define software testing standards and quality control process

SOFTWARE TESTING PRINCIPLES

1: Complete testing is impossible.

2: Software testing is not simple.

Reasons:

- Quality testing requires testers to understand a system/product completely
- Quality testing needs adequate test set, and efficient testing methods
- A very tight schedule and lack of test tools.

3: Testing is risk-based.

4: Testing must be planned.

5: Testing requires independence.

6: Quality software testing depends on:

- Good understanding of software products and related domain application
- Cost-effective testing methodology, coverage, test methods, and tools.
- Good engineers with creativity, and solid software testing experience